

KAVI MOOKHERJEE AMODT

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EDUCATION AND COURSEWORK

UC Berkeley

Until 2026

Computer Science, Cognitive Science; Linguistics Minor

3.72 GPA

CS: Probability and Random Processes (EECS 126), Artificial Intelligence (CS 188), Algorithms (CS 170), Computer Security (CS 161), Software Engineering (CS 169), Algorithmic Economics (CS 177), Computability and Complexity (CS 172), Econometrics (Econ 141), Computer Architecture (CS 61C), Data Structures (CS 61B), Discrete Math and Probability (CS 70), Linear Algebra (CS 56), Calculus (51/52), Statistics (2)

PROJECTS

Secure File Sharing System

- Designed and implemented a cryptographically secure file sharing system in Go. Built from scratch; no vibe coding.
- Tested in a fully adversarial server environment where adversaries have full write access to datastore and read access to keystore.

Regression and Gradient Descent From Scratch

- Linear and logistic regression from first principles. Gradient descent and closed form solvers.
- Added ridge (L2) and lasso (L1) regularization and tuned penalty strength via k-fold cross-validation, using bias-variance analysis to diagnose over- and underfitting.

Assembly Level Digit Classifier

- Implemented a neural network inference engine in RISC-V assembly, capable of classifying handwritten digits through matrix operations and ReLU activation.

MCMC Cipher Decoder

- Implemented the Metropolis-Hastings from scratch in Python/NumPy.
- Sample from arbitrary target distributions, then applied it to decrypt substitution ciphers by treating the space of key permutations as a Markov chain and scoring candidate keys against an English-language bigram frequency matrix built from War and Peace.

Prediction Market Simulator

- Simulating and visualizing prediction markets with LMSR online learning algorithm in C++.

EXPERIENCE

Software Engineering Intern

2024

Tools for Humanity

San Francisco, CA

- Wrote Rust, Python, and Bash for an iris scanning device called the Orb.
- Built Rust CLI tools for accessing Orb metrics.

Research Assistant

2025–

Language and Cognitive Development Lab

UC Berkeley

- Built experiment for word-object learning models using dynamic field theory in MATLAB.
- Wrote data processing scripts in Python.

Research Assistant

2021–2023

UCSC Linguistics Department

Santa Cruz, CA

- Eyetracking research on underspecification, parsing of syntactic ambiguities.
- Operated eye-tracker, worked with participants, collected data.
- Worked with Amanda Rysling (UCSC), Jack Duff (UCLA).

SKILLS

Languages: Python, Go, C++, C, Java, R, RISC-V Assembly, SQL, Bash

Libraries & Frameworks: PyTorch, NumPy, SciPy, pandas, Matplotlib

Tools & Platforms: Git, Jupyter, Linux, Model Context Protocol (MCP)

Domains: Machine learning, applied cryptography, information theory, probabilistic modeling, computer architecture